Assignment 6

Question 1

Create the following data set with 20 different years and perform the following operations using „dplyr" library.

Solution

```
# Question 1: Create the following data set with 20 different years and perform the
following operations using „dplyr" library.
df <- data.frame(
  country = rep(c("India", "USA", "China", "Brazil", "UK", "Germany", "France", "Italy",
"Canada", "Australia",
                  "Japan", "South Korea", "Mexico", "South Africa", "Russia",
                  "Spain", "Netherlands", "Sweden", "Norway", "Denmark", "Finland",
"Poland", "Portugal",
                  "Argentina", "Chile", "Colombia", "New Zealand", "Thailand", "Vietnam",
"Philippines"), each = 2),
  continent = rep(c("Asia", "North America", "Asia", "South America", "Europe", "Europe",
"Europe", "Europe", "North America", "Oceania",
                    "Asia", "Asia", "North America", "Africa", "Europe",
                    "Europe", "Europe", "Europe", "Europe", "Europe", "Europe", "Europe",
"Europe",
                    "South America", "South America", "South America", "Oceania", "Asia",
"Asia", "Asia"), each = 2),
  year = rep(2000:2014, times = 2),
  lifeExp = runif(60, 60, 85),
  pop = sample(1e6:1.4e9, 60, replace = TRUE),
  gdpPerc = sample(3000:60000, 60, replace = TRUE)
)

print(summary(df))
```

Output

```
country            continent              year          lifeExp
 Length:60          Length:60          Min.   :2000   Min.   :60.26
 Class :character   Class :character   1st Qu.:2003   1st Qu.:68.36
 Mode  :character   Mode  :character   Median :2007   Median :72.52
                                       Mean   :2007   Mean   :73.13
                                       3rd Qu.:2011   3rd Qu.:78.68
                                       Max.   :2014   Max.   :84.47
      pop               gdpPerc
 Min.   :8.012e+06   Min.   : 4276
 1st Qu.:2.126e+08   1st Qu.:14291
 Median :6.008e+08   Median :26167
 Mean   :6.287e+08   Mean   :28973
 3rd Qu.:1.007e+09   3rd Qu.:41720
 Max.   :1.313e+09   Max.   :58945
```

Question 1. (A)

How many unique countries per continent.

Solution

```r
library(dplyr)

unique_countries <- select(df, country, continent)
unique_countries <- unique(unique_countries)

print(count(group_by(unique_countries, continent)))
```

Output

```
A tibble: 6 × 2
continent     unique_countries
Africa                       1
Asia                         7
Europe                      13
North America                3
Oceania                      2
South America                4
```

Question 1. (b)

Which European nation had the lowest GDP per capita in a given year?

Solution

```r
lowest_gdp_europe <- df %>%
  filter(continent == "Europe") %>%
  group_by(year) %>%
  select(country, year, gdpPerc) %>%
  arrange(gdpPerc) %>%
  slice(1)

print(lowest_gdp_europe)
```

Output

```
A tibble: 15 × 3
Groups:   year [15]
country      year gdpPerc
 1 Spain        2000   □[4m2□[24m□[4m5□[24m395
 2 Spain        2001   □[4m4□[24m□[4m1□[24m012
 3 Netherlands  2002   □[4m5□[24m□[4m3□[24m617
 4 Netherlands  2003   □[4m1□[24m□[4m0□[24m773
 5 Sweden       2004   □[4m1□[24m□[4m9□[24m775
 6 Sweden       2005   □[4m4□[24m□[4m9□[24m475
 7 Norway       2006   □[4m3□[24m□[4m0□[24m474
 8 Norway       2007   □[4m3□[24m□[4m1□[24m538
 9 Denmark      2008    □[4m7□[24m364
10 UK           2009   □[4m2□[24m□[4m1□[24m542
11 Germany      2010   □[4m3□[24m□[4m6□[24m682
12 Finland      2011    □[4m8□[24m264
13 France       2012    □[4m4□[24m709
14 France       2013   □[4m1□[24m□[4m6□[24m364
15 Italy        2014   □[4m1□[24m□[4m7□[24m723
```

QUESTION 1. (C)

According to the data available, what was the average life expectancy across each continent in a given year?

SOLUTION

```
avg_life_exp <- aggregate(lifeExp ~ continent + year, data = df, FUN = mean)
print(avg_life_exp)
```

OUTPUT

```
continent year  lifeExp
1            Asia 2000 61.97577
2          Europe 2000 69.73997
3            Asia 2001 65.38429
4          Europe 2001 75.36298
5   North America 2001 77.48323
6   South America 2001 83.31628
7          Europe 2002 78.64270
8   North America 2002 72.60816
9   South America 2002 80.81569
10         Europe 2003 72.60430
11 North America 2003 80.13556
12        Oceania 2003 60.84253
13 South America 2003 60.26338
14           Asia 2004 63.39555
15         Europe 2004 82.48371
16        Oceania 2004 71.68405
17 South America 2004 76.44950
18           Asia 2005 81.82738
19         Europe 2005 69.22234
20 South America 2005 68.97537
21           Asia 2006 76.79630
22         Europe 2006 83.98541
23 South America 2006 66.72524
24           Asia 2007 72.76388
25         Europe 2007 69.83252
26        Oceania 2007 80.69767
27 South America 2007 70.18291
28           Asia 2008 73.26076
29         Europe 2008 70.36375
30        Oceania 2008 84.22953
31           Asia 2009 83.46185
32         Europe 2009 71.79938
33 North America 2009 75.90216
34           Asia 2010 66.78098
35         Europe 2010 68.22478
36 North America 2010 78.77737
37         Africa 2011 82.84786
38           Asia 2011 70.19840
39         Europe 2011 69.80194
40         Africa 2012 73.87304
41           Asia 2012 70.27315
42         Europe 2012 69.91503
43           Asia 2013 76.18017
44         Europe 2013 73.02079
45           Asia 2014 70.65546
46         Europe 2014 75.84524
```

Question 1. (d)

What 5 countries have the highest total GDP over all years combined?

Solution

```
gdp_df <- df %>%
  mutate(total_gdp = as.numeric(gdpPerc) * as.numeric(pop))

top_5_countries <- aggregate(total_gdp ~ country, data = gdp_df, FUN = sum) %>%
  arrange(desc(total_gdp)) %>%
  head(5)
print(top_5_countries)
```

Output

```
country     total_gdp
1       Russia 1.437916e+14
2       Mexico 8.652847e+13
3       Sweden 7.355758e+13
4        India 6.234923e+13
5 Netherlands 5.535939e+13
```

Question 1. (e)

What countries and years had life expectancies of at least 80 years?

Solution

```
high_life_exp <- df %>%
  filter(lifeExp >= 80) %>%
  select(country, year, lifeExp)

print(high_life_exp)
```

Output

```
        country year  lifeExp
1           USA 2003 80.13556
2        Canada 2002 80.96746
3         Japan 2005 84.46835
4  South Africa 2011 82.84786
5        Russia 2014 83.78255
6        Sweden 2004 82.48371
7        Norway 2006 83.98541
8     Argentina 2001 83.31628
9     Argentina 2002 80.81569
10  New Zealand 2007 80.69767
11  New Zealand 2008 84.22953
12     Thailand 2009 83.46185
```

Question 1. (f)

What 10 countries have the strongest correlation (in either direction) between life expectancy and per capita GDP?

Solution

```
df <- mutate(df, gdpCorr = cor(gdpPerc, lifeExp))
df <- group_by(df, country) %>%
      arrange(desc(abs(gdpCorr))) %>%
      slice(1)
print(tail(df, 10))
```

Output

```
   country      continent      year lifeExp        pop gdpPerc gdpCorr
 1 Portugal     Europe         2014    79.6 1106468148   34667   0.112
 2 Russia       Europe         2013    82.5  203307819   21899   0.112
 3 South Africa Africa         2011    74.1  642006400   23944   0.112
 4 South Korea  Asia           2007    79.8  298705609   43262   0.112
 5 Spain        Europe         2000    84.1  856771125   49578   0.112
 6 Sweden       Europe         2004    78.4  384053295   41039   0.112
 7 Thailand     Asia           2009    66.9  229008478   20186   0.112
 8 UK           Europe         2008    66.1  909143577   51218   0.112
 9 USA          North America  2002    64.5  882120679   36360   0.112
 10 Vietnam      Asia           2011    72.3   58478689   17796   0.112
```

Question 1. (g)

Which combinations of continent (besides Asia) and year have the highest average population across all countries?

Solution

```
mutate(gdp_df, corr = lifeExp/gdpPerc)
top_10_corr <- aggregate(corr ~ country, data = gdp_df, FUN = function(x) cor(x, use =
"complete.obs")) %>%
  arrange(desc(abs(corr))) %>%

print(head(top_10_corr, 10))
```

Output

```
country      continent     year lifeExp   pop        gdpPerc total_gdp
1   India        Asia          2000 61.97577 1271264396 42184   5.362702e+13
2   India        Asia          2001 65.38429  499153698 17474   8.722212e+12
3   USA          North America 2002 64.24885  537490731 53220   2.860526e+13
4   USA          North America 2003 80.13556  246308518 15541   3.827881e+12
5   China        Asia          2004 63.39555  135614236 35285   4.785148e+12
6   China        Asia          2005 79.18640  943832654 25288   2.386764e+13
7   Brazil       South America 2006 68.16280 1199360105  6157   7.384460e+12
8   Brazil       South America 2007 70.18291  712399179 37798   2.692726e+13
9   UK           Europe        2008 74.63302  478771350 16002   7.661299e+12
10  UK           Europe        2009 68.21730  312664356 21542   6.735416e+12
11  Germany      Europe        2010 64.23460  451188843 36682   1.655051e+13
12  Germany      Europe        2011 67.17170 1096698682  9669   1.060398e+13
13  France       Europe        2012 63.65903   15747597  4709   7.415543e+10
14  France       Europe        2013 70.96158  425105885 16364   6.956433e+12
15  Italy        Europe        2014 71.83112  213793570 17723   3.789063e+12
16  Italy        Europe        2000 70.38039 1313363207 26939   3.538069e+13
17  Canada       North America 2001 77.48323  208869615 32465   6.780952e+12
18  Canada       North America 2002 80.96746 1109633463  8316   9.227712e+12
19  Australia    Oceania       2003 60.84253   70360914 12798   9.004790e+11
20  Australia    Oceania       2004 71.68405 1033056924 19899   2.055680e+13
```

```
21 Japan        Asia          2005 84.46835  655426026 53910   3.533402e+13
22 Japan        Asia          2006 76.79630  874776642 10289   9.000577e+12
23 South Korea  Asia          2007 72.76388  340780719 25249   8.604372e+12
24 South Korea  Asia          2008 73.26076  120173370 33618   4.039988e+12
25 Mexico       North America 2009 75.90216 1241919660 56900   7.066523e+13
26 Mexico       North America 2010 78.77737 1233821057 12857   1.586324e+13
27 South Africa Africa        2011 82.84786  160204427 56702   9.083911e+12
28 South Africa Africa        2012 73.87304  597670713 36393   2.175103e+13
29 Russia       Europe        2013 68.40803 1269887934 55844   7.091562e+13
30 Russia       Europe        2014 83.78255 1250660510 58270   7.287599e+13
31 Spain        Europe        2000 72.73875  998222050 25395   2.534985e+13
32 Spain        Europe        2001 75.36298  120919878 41012   4.959166e+12
33 Netherlands  Europe        2002 78.64270  955930590 53617   5.125413e+13
34 Netherlands  Europe        2003 72.60430  381069521 10773   4.105262e+12
35 Sweden       Europe        2004 82.48371  603973317 19775   1.194357e+13
36 Sweden       Europe        2005 69.22234 1245356331 49475   6.161400e+13
37 Norway       Europe        2006 83.98541 1281119230 30474   3.904083e+13
38 Norway       Europe        2007 69.83252   20089523 31538   6.335834e+11
39 Denmark      Europe        2008 66.09448    8011521  7364   5.899684e+10
40 Denmark      Europe        2009 75.38146  137550455 32902   4.525685e+12
41 Finland      Europe        2010 72.21495  328691868 56739   1.864965e+13
42 Finland      Europe        2011 72.43218  642156409  8264   5.306781e+12
43 Poland       Europe        2012 76.17103  901332160 43732   3.941706e+13
44 Poland       Europe        2013 79.69277  155956502 19762   3.082012e+12
45 Portugal     Europe        2014 71.92204  193366864 58526   1.131699e+13
46 Portugal     Europe        2000 66.10076   76890520 58945   4.532312e+12
47 Argentina    South America 2001 83.31628 1065119317 45590   4.855879e+13
48 Argentina    South America 2002 80.81569 1145267526  5135   5.880949e+12
49 Chile        South America 2003 60.26338  794797340 41565   3.303575e+13
50 Chile        South America 2004 76.44950  727250123 11970   8.705184e+12
51 Colombia     South America 2005 68.97537  812139894 11843   9.618173e+12
52 Colombia     South America 2006 65.28769  116880593 16960   1.982295e+12
53 New Zealand  Oceania       2007 80.69767 1213026725 14769   1.791519e+13
54 New Zealand  Oceania       2008 84.22953  865717428 17507   1.515612e+13
55 Thailand     Asia          2009 83.46185  400162831 40355   1.614857e+13
56 Thailand     Asia          2010 66.78098  519265232 40129   2.083759e+13
57 Vietnam      Asia          2011 70.19840  616056518  7588   4.674637e+12
58 Vietnam      Asia          2012 70.27315  499926939 30748   1.537175e+13
59 Philippines  Asia          2013 76.18017   29784803 45594   1.358008e+12
60 Philippines  Asia          2014 70.65546  847957709  4276   3.625867e+12
   corr
1  0.001469177
2  0.003741804
3  0.001207231
4  0.005156397
5  0.001796671
6  0.003131382
7  0.011070781
8  0.001856789
9  0.004663980
10 0.003166711
11 0.001751120
12 0.006947120
13 0.013518588
14 0.004336445
15 0.004052989
16 0.002612584
17 0.002386670
18 0.009736347
19 0.004754065
20 0.003602395
```

```
21 0.001566840
22 0.007463923
23 0.002881852
24 0.002179212
25 0.001333957
26 0.006127197
27 0.001461110
28 0.002029869
29 0.001224984
30 0.001437833
31 0.002864294
32 0.001837584
33 0.001466749
34 0.006739469
35 0.004171110
36 0.001399138
37 0.002755969
38 0.002214234
39 0.008975351
40 0.002291091
41 0.001272757
42 0.008764785
43 0.001741769
44 0.004032627
45 0.001228890
46 0.001121397
47 0.001827512
48 0.015738207
49 0.001449859
50 0.006386758
51 0.005824147
52 0.003849510
53 0.005463990
54 0.004811192
55 0.002068191
56 0.001664157
57 0.009251239
58 0.002285454
59 0.001670838
60 0.016523728
```

Question 1. (h)

Which three countries have had the most consistent population estimates (i.e. lowest standard deviation) across the years of available data?

Solution

```
consistent_pop <- aggregate(pop ~ country, data = df, FUN = sd) %>%
  arrange(pop) %>%
  head(3)

print(consistent_pop)
```

Output

```
country      pop
1  Mexico  5726577
```

```
2  Russia 13595842
3   Chile 47763095
```

Question 1. (i)

Excluding records from a given year, which observations indicate that the population of a country has decreased from the previous year and the life expectancy has increased?

Solution

```
pop_life_change <- df %>%
  arrange(country, year) %>%
  mutate(pop_change = pop - lag(pop),
   lifeExp_change = lifeExp - lag(lifeExp)) %>%
  filter(pop_change < 0 & lifeExp_change > 0) %>%
  select(country, year, pop, lifeExp)

print(pop_life_change)
```

Output

```
country year        pop  lifeExp
1       Brazil 2007  712399179 70.18291
2       Canada 2001  208869615 77.48323
3        Chile 2004  727250123 76.44950
4      Denmark 2008    8011521 66.09448
5        India 2001  499153698 65.38429
6        Italy 2014  213793570 71.83112
7       Mexico 2010 1233821057 78.77737
8  New Zealand 2008  865717428 84.22953
9       Poland 2013  155956502 79.69277
10      Russia 2014 1250660510 83.78255
11 South Korea 2008  120173370 73.26076
12       Spain 2001  120919878 75.36298
13    Thailand 2009  400162831 83.46185
14          UK 2008  478771350 74.63302
15         USA 2003  246308518 80.13556
16     Vietnam 2012  499926939 70.27315
```

Question 2

Create a database file “DataSet.csv” that contains 10 records of medicine with attribute : MedID, Med_Name, Company, Manf_year, Exp_date, Quantity_in_stock, Sales.

Solution

```
df <- data.frame(
  medid = 1:10,
  med_name = c("meda", "medb", "medc", "medd", "mede", "medf", "medg", "medh", "medi",
"medj"),
  company = c("comp1", "comp2", "comp1", "comp3", "comp2", "comp4", "comp3", "comp1",
"comp4", "comp2"),
  manf_year = c(2018, 2019, 2018, 2020, 2019, 2021, 2020, 2018, 2021, 2019),
  exp_date = as.Date(trimws(iconv(c("2023-12-31", "2024-06-30", "2023-11-30",
```

```
                                      "2025-01-31", "2024-05-31", "2026-12-31",
                                      "2025-03-31", "2023-10-31", "2026-11-30",
"2024-07-31"),
                                    from = "", to = "UTF-8"))),
  quantity_in_stock = c(100, 150, 200, 120, 180, 160, 140, 130, 170, 110),
  sales = c(5000, 7000, 6000, 8000, 7500, 9000, 8500, 6500, 9500, 7200)
)
print(df)
write.csv("DataSet.csv", row.names=FALSE)
```

Output

```
medid med_name company manf_year   exp_date quantity_in_stock sales
1       1      meda    comp1      2018 2023-12-31              100   5000
2       2      medb    comp2      2019 2024-06-30              150   7000
3       3      medc    comp1      2018 2023-11-30              200   6000
4       4      medd    comp3      2020 2025-01-31              120   8000
5       5      mede    comp2      2019 2024-05-31              180   7500
6       6      medf    comp4      2021 2026-12-31              160   9000
7       7      medg    comp3      2020 2025-03-31              140   8500
8       8      medh    comp1      2018 2023-10-31              130   6500
9       9      medi    comp4      2021 2026-11-30              170   9500
10     10      medj    comp2      2019 2024-07-31              110   7200
"x"
"DataSet.csv"
```

Question 2. (A)

Read the data file and show the first 4 record of the file.

Solution

```
data <- read.csv("DataSet.csv")
print(head(data, 4))
```

Output

```
MedID Med_Name Company Manf_year   Exp_date Quantity_in_stock Sales
1      1      MedA    Comp1      2018  2023-12-31              100   5000
2      2      MedB    Comp2      2019  2024-06-30              150   7000
3      3      MedC    Comp1      2018  2023-11-30              200   6000
4      4      MedD    Comp3      2020  2025-01-31              120   8000
```

Question 2. (B)

Read the data file and show the last 4 record of the file.

Solution

```
print(tail(data, 4))
```

Output

```
MedID Med_Name Company Manf_year   Exp_date Quantity_in_stock Sales
7       7      MedG    Comp3      2020 2025-03-31              140   8500
8       8      MedH    Comp1      2018 2023-10-31              130   6500
```

```
9      9      MedI   Comp4      2021 2026-11-30             170  9500
10     10     MedJ   Comp2      2019 2024-07-31             110  7200
```

Question 2. (c)

Find the coorelation between Quantity_in_stock and Exp_date.

Solution

```
data$Exp_date <- as.numeric(data$Exp_date)
data <- na.omit(data)
correlation <- cor(data$Quantity_in_stock, data$Exp_date)
print(correlation)
```

Output

```
[1] 0.2084504
```

Question 2. (d)

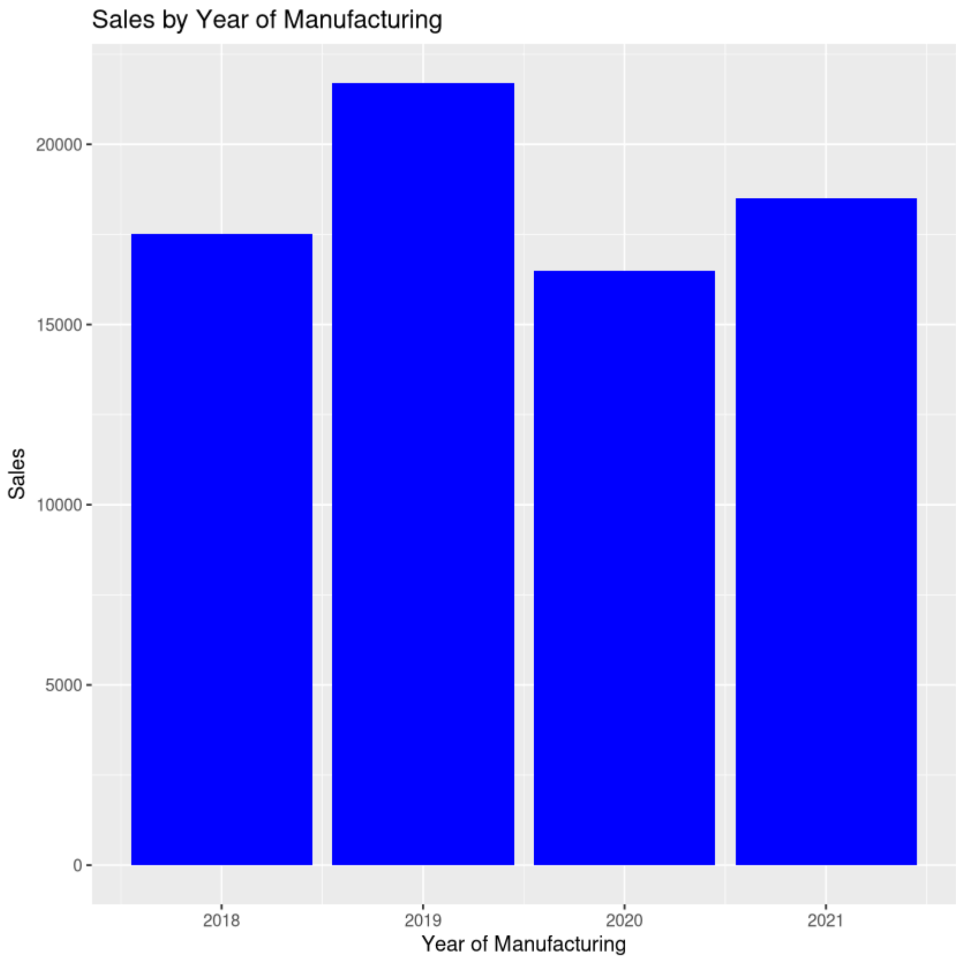
Plot the bar graph for the Sales with year of manufacturing.

Solution

```
library(ggplot2)
ggplot(data, aes(x = Manf_year, y = Sales)) +
geom_bar(stat = "identity", fill = "blue") +
labs(
  title = "Sales by Year of Manufacturing",
  x = "Year of Manufacturing",
  y = "Sales"
)
```

Output

```
plot without title
```



Question 2. (e)

Find the company having more than one type of medicine.

Solution

```
company_medicine <- aggregate(Med_Name ~ Company, data = data, FUN = function(x)
length(unique(x)))
print(company_medicine[company_medicine$Med_Name > 1, ])
```

Output

```
Company Med_Name
1    Comp1        3
2    Comp2        3
3    Comp3        2
4    Comp4        2
```

Question 2. (f)

Find the type of Medicine available.

Solution

```
medicine_types <- unique(data$Med_Name)
print(medicine_types)
```

Output

```
[1] "MedA" "MedB" "MedC" "MedD" "MedE" "MedF" "MedG" "MedH" "MedI" "MedJ"
```

Question 2. (g)

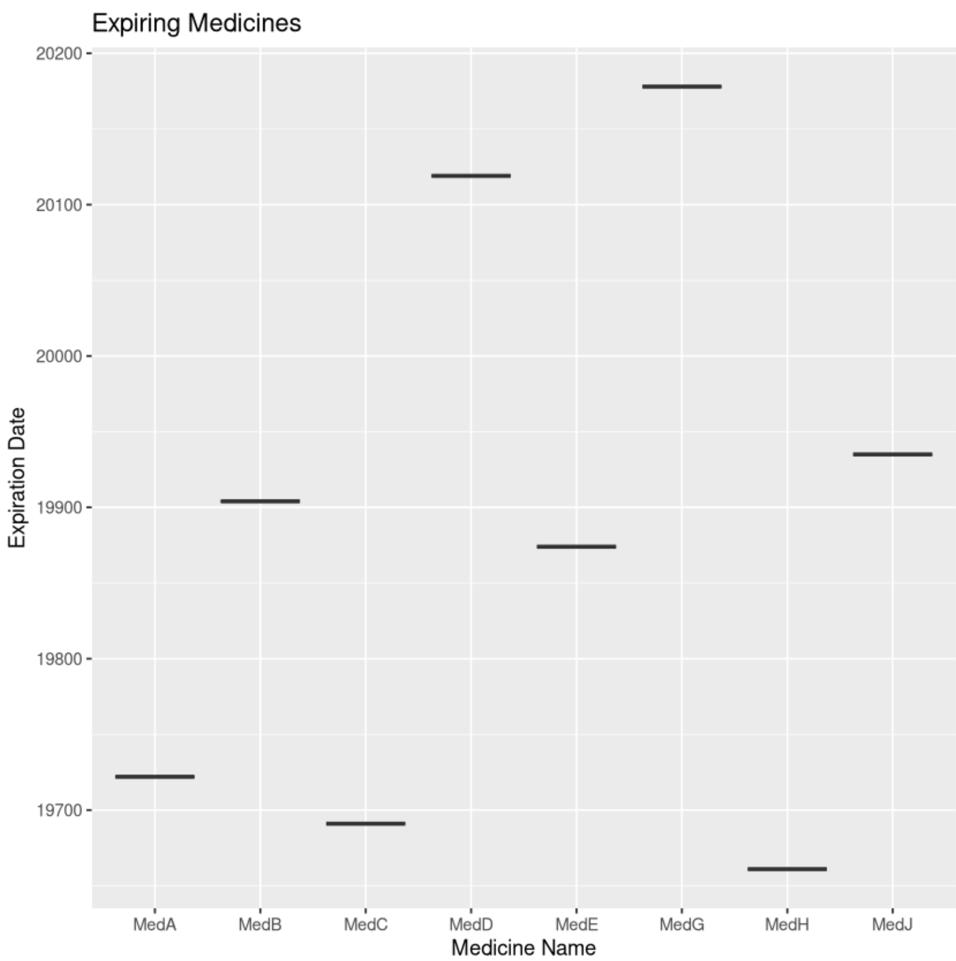
Which medicines are expiring? Show by box plots.

Solution

```
expiring_medicines <- data[data$Exp_date < Sys.Date(), ]
ggplot(expiring_medicines, aes(x = Med_Name, y = Exp_date)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Expiring Medicines", x = "Medicine Name", y = "Expiration Date")
```

Output

```
plot without title
```



Question 2. (h)

Find the average stock in the store.

Solution

```
average_stock <- mean(data$Quantity_in_stock)
print(average_stock)

average_stock_per_med <- aggregate(Quantity_in_stock ~ Med_Name, data = data, FUN = mean)
print(average_stock_per_med)
```

Output

```
[1] 146
   Med_Name Quantity_in_stock
1      MedA               100
2      MedB               150
3      MedC               200
4      MedD               120
5      MedE               180
6      MedF               160
7      MedG               140
8      MedH               130
9      MedI               170
10     MedJ               110
```

Question 2. (i)

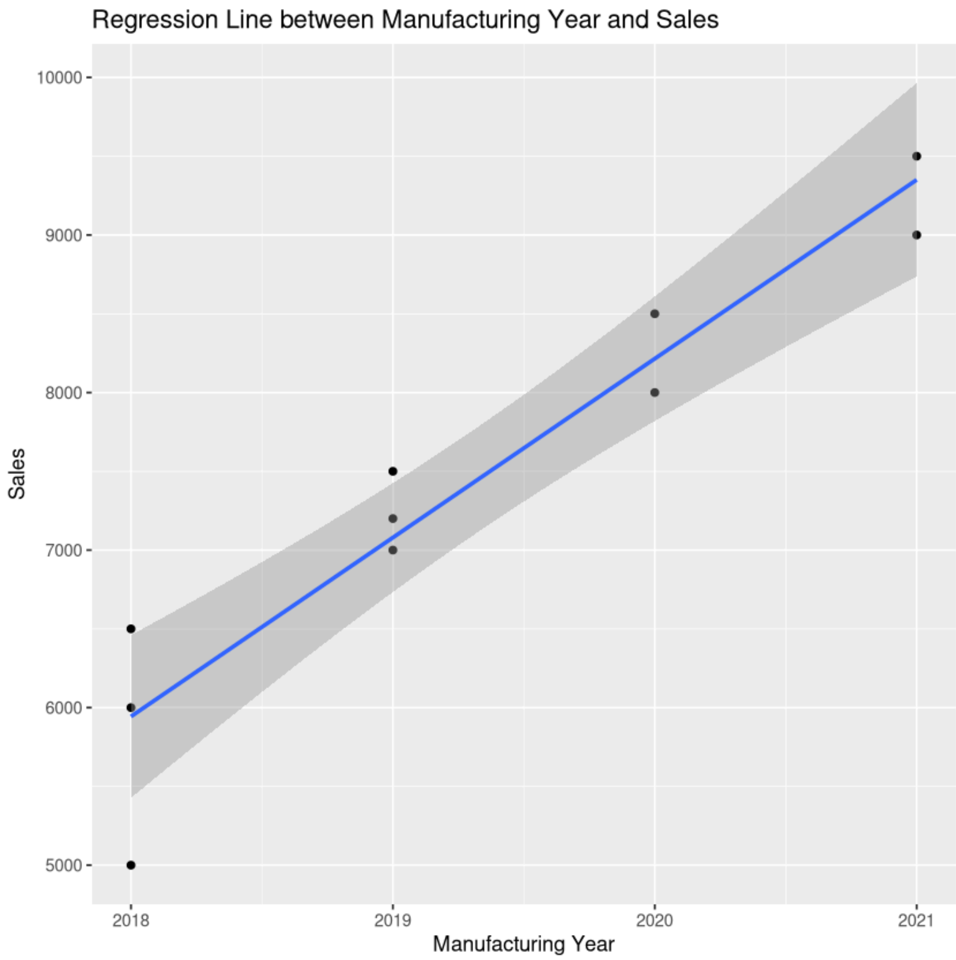
Draw the regression line between Manufacturing year and Sales.

Solution

```
ggplot(data, aes(x = Manf_year, y = Sales)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "Regression Line between Manufacturing Year and Sales", x = "Manufacturing
Year", y = "Sales")
```

Output

```
[1m[22m`geom_smooth()` using formula = 'y ~ x'
plot without title
```


Regression Line between Manufacturing Year and Sales
10000
9000
8000
7000
6000
5000
Sales
2018
2019
2020
2021
Manufacturing Year